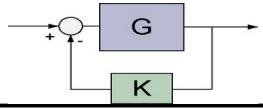
HW2 Review



Inverse Laplace w/ Complex Roots

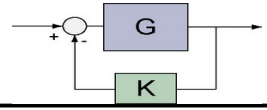
- Using MATLAB `residue` function and handout to interpret

Transfer function

- From ODEs
- Using MATLAB

USV Model

Notes

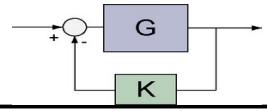


Please put your name on your assignment.

MATLAB

- Using MATLAB Copilot?

Observation



$\delta(\mathbf{t}) = .3$

$$R(s) = \frac{.3}{s} * \frac{2.5}{s+1.5} = \frac{.75}{s(s+1.5)}$$

At steady state t approaches infinity meaning s approaches 0. Therefore to get a steady state value we can plug 0 into our function above for S and solve. <u>This gives us a steady state value of .5.</u>

Assignment #2

2.8. $\frac{d^3y}{dt^3} + 3\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + y = \frac{d^3x}{dt^3} + 4\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x$

assume all initial values are (4) so I'm going to ignore/not write it

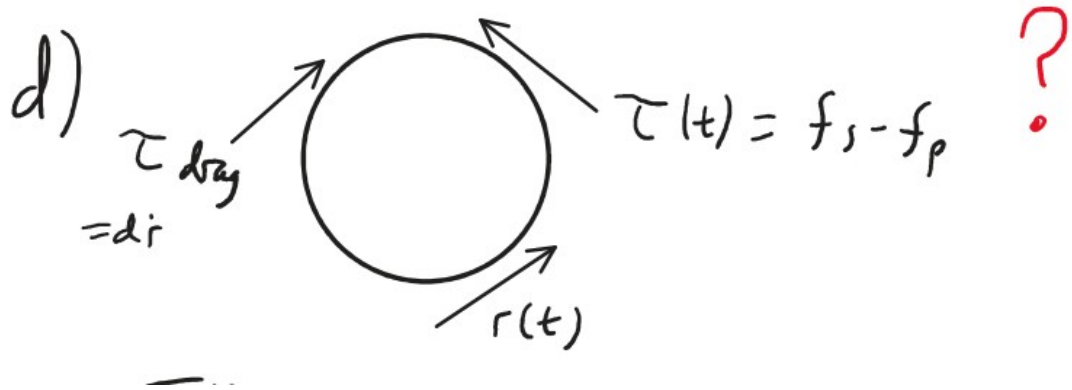
$$s^3Y(s) + 3s^2Y(s) + 5sY(s) + Y(s) = s^3X(s) + 4s^2X(s) + 6sX(s) + 8X(s)$$

$$Y(s)(s^3 + 3s^2 + 5s + 1) = X(s)(s^3 + 4s^2 + 6s + 8)$$

$$\Rightarrow \frac{Y(s)}{X(s)} = \frac{s^3 + 4s^2 + 6s + 8}{s^3 + 3s^2 + 5s + 1}$$

USV Yaw Model

Free Body Diagram

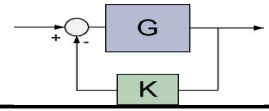


Yaw vs Yaw-Rate

$$\frac{\dot{r}}{\tau} = ?! = \frac{\omega = \dot{r} = s(r)}{\tau} = s\left(\frac{r}{\tau}\right)$$

← need to review this process again!





Previous Quarters

b.) $G(s) = \frac{2}{s(s^2+s+6.25)}$

$K = -0.0615, \quad 6.1231$

$\theta = -0.0077, \quad 0$

$a = 0.5, \quad 0$

$\omega = 4, \quad 0$

$f(t) = 2[-0.0615e^{-0.5t}\cos(4t) + (-0.077)e^{-.5t}\sin(4t)] + 0.1231$?

↳ on a), I accidentally didn't do 2x the rest. For this one, I was about to and then realized I didn't do it in the first part(b). Why don't we?

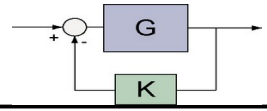
↳ $F(s) = \frac{0.1231}{s+(0+0)} + \frac{0.1231}{s+0} = 2\left[\frac{K}{s}\right] = \frac{2K}{s} \Rightarrow \frac{2(0.1231)}{s}$

solutions don't do this. Not sure which is $\mathcal{L}^{-1}(\quad)$ → 0.2462

wrong & assume it's me.

Seems like we should have 2K on the final part of $f(t)$.

PFE



Description

`[r,p,k] = residue(b,a)` finds the residues, poles, and direct term of a Partial Fraction Expansion of the ratio of two polynomials, where the expansion is of the form

$$\frac{b(s)}{a(s)} = \frac{b_m s^m + b_{m-1}s^{m-1} + \ldots + b_1 s + b_0}{a_n s^n + a_{n-1}s^{n-1} + \ldots + a_1 s + a_0} = \frac{r_n}{s - p_n} + \ldots + \frac{r_2}{s - p_2} + \frac{r_1}{s - p_1} + k(s).$$

The inputs to `residue` are vectors of coefficients of the polynomials `b = [bm ... b1 b0]` and `a = [an ... a1 a0]`. The outputs are the residues `r = [rn ... r2 r1]`, the poles `p = [pn ... p2 p1]`, and the polynomial `k`. For most textbook problems, `k` is `0` or a constant.

G(s)

```
num = 2;
den = [1 1 16.25 0];
[R2,P2,K2] = residue(num,den)
```

$$a = 0.5, \omega = 4, \alpha = -0.0615, \beta = -0.0077$$

$$G(s) = 2\left[\frac{-0.0615(s + 0.5) - 0.0077(4)}{(s + 0.5)^2 + 4^2}\right] + \frac{0.1231}{s}$$

$$g(t) = 2e^{-0.5t}[-0.0615cos(4t) - 0.0077sin(4t)] + 0.1231$$

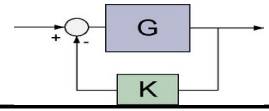
```
R2 = 3×1 complex
   -0.0615 + 0.0077i
   -0.0615 - 0.0077i
    0.1231 + 0.0000i

P2 = 3×1 complex
   -0.5000 + 4.0000i
   -0.5000 - 4.0000i
    0.0000 + 0.0000i

K2 =
     []

R3 = 4×1 complex
```

PFE: Repeated Roots



If `p(j) = ... = p(j+m-1)` is a pole of multiplicity `m`, then the expansion includes terms of the form

$$\frac{r_j}{s-p_j}+\frac{r_{j+1}}{(s-p_j)^2}+\ldots+\frac{r_{j+m-1}}{(s-p_j)^m}.$$

$G(s)$

```
num = 42;
den = [1 7 16 12];
G = tf(num, den)
[r, p, k] = residue(num, den)
```

$$G(s)=\frac{42}{s^3+7s^2+16s+12}=42\left[\frac{1}{s+3}-\frac{1}{s+2}+\frac{1}{(s+2)^2}\right]$$

$$g(t)=42[e^{-3t}-e^{-2t}+e^{-2t}t]=42[e^{-3t}+e^{-2t}(t-1)]$$

```
G =

             42
  ----------------------
  s^3 + 7 s^2 + 16 s + 12

Continuous-time transfer function.
Model Properties

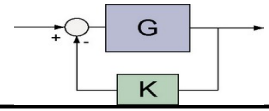
r = 3×1

    42.0000
   -42.0000
    42.0000

p = 3×1

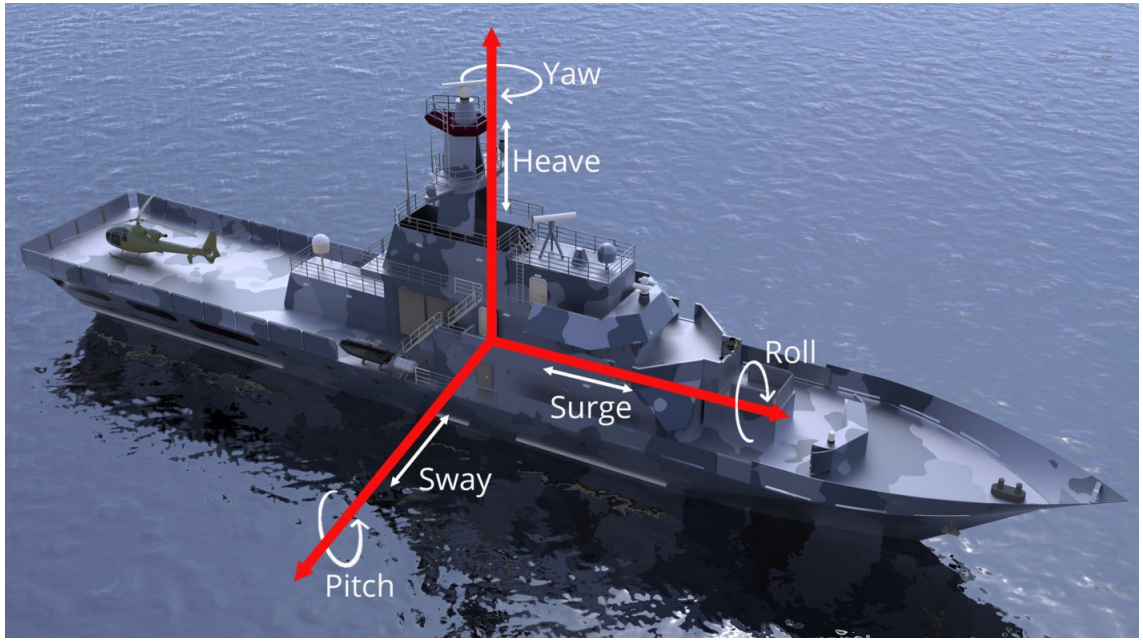
    -3.0000
    -2.0000
    -2.0000
```

MATLAB and Transfer Functions



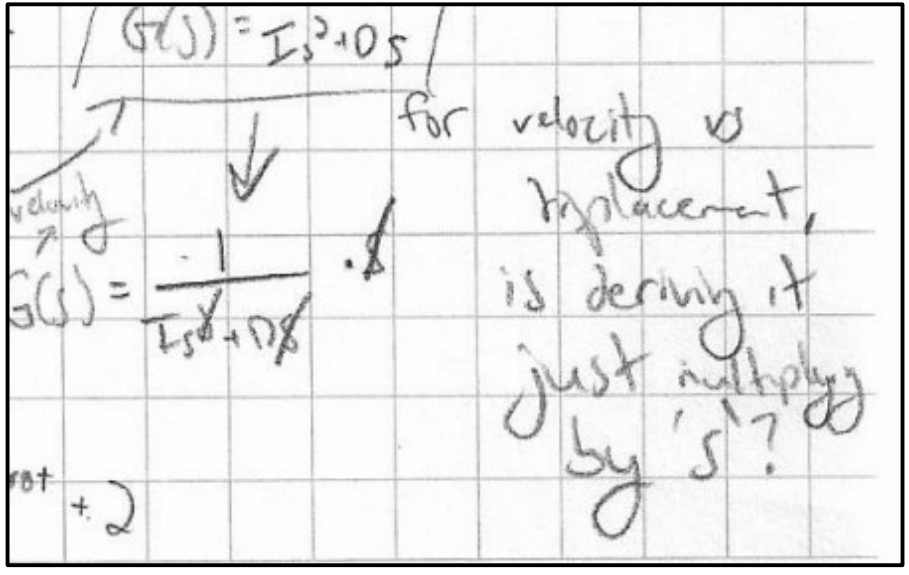

```
%% Ratio of factors
z = [-15 -26 -72];
p = [0 -55 roots([1 5 30])' -56 roots([1 27 52])']; %definitely forgot to
%add the ' to the roots functions when I did it before looking at the
%solution.
k = 5;
G = zpk(z,p,k)
%% Ratio of polynomials
num = 5*conv(conv([1 15],[1 26]),[1 72]);
den = [conv([1 27 52],conv([1 56],conv([1 5 30],conv([1 0],[1 55]))))];% this i
tf(num,den)
```

USV Model

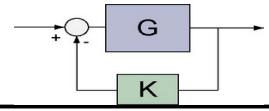


$$G_{\text{surge}}(s) = \frac{\text{forward velocity}}{\text{thrust}} = \frac{1}{Ms + D}$$

$$G_{\text{yaw}}(s) = \frac{\text{angular displacement}}{\text{torque}} = \frac{1}{s(Is + d)}$$

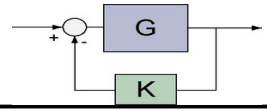






Previous Quarters

USV Model Response

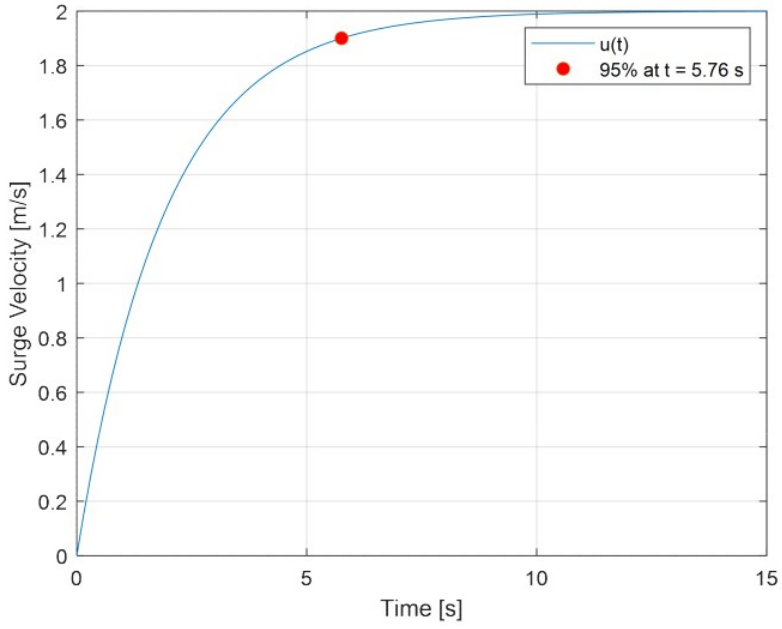


b) $M(sU(s) - u(0)) + DU(s) = \frac{37.5}{s}$

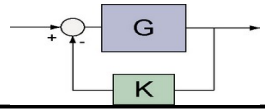
$$U(s)[M_s + D] = \frac{37.5}{s}$$

$$U(s) = \frac{37.5}{(M_s + D)s} = \frac{1}{M_s + D} \cdot \frac{37.5}{s}$$

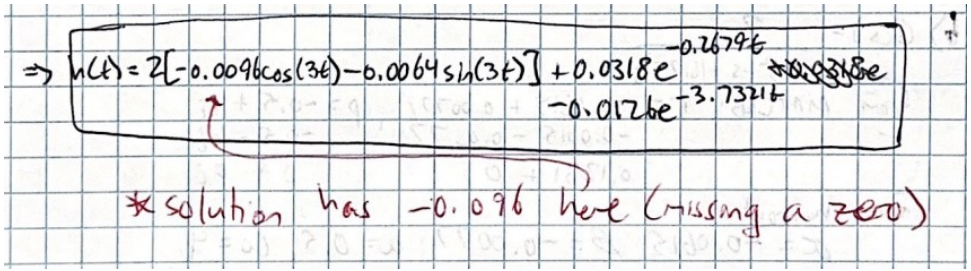
Review?



HW2 Review



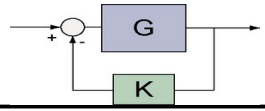
- Some are grading themselves – doing the HW and then grading with a red pen. That is a fantastic process. Well done!
  - Found a typo in solutions.



- Make sure you are keeping up – if you are falling behind, let me know.

- MATLAB
  - Not required to submit MATLAB (live) script
  - Get in the habit of using a (live) script – not the command window.
  - Take the time to learn plotting syntax

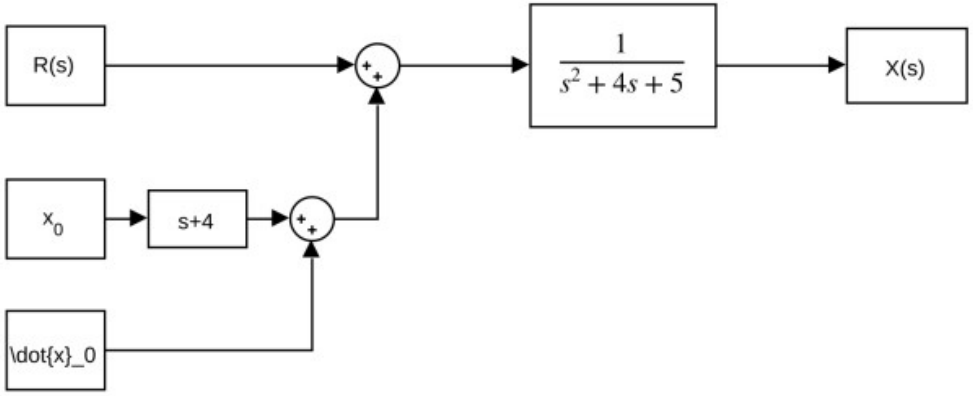
Nise 2.12



A system is described by the following differential equation: [Section 2.3]

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 1$$

with the initial conditions $x(0) = 1$, $\dot{x}(0) = -1$. Show a block diagram of the system, giving its transfer function and all pertinent inputs and outputs. (Hint: the initial



Given differential equation

$$\ddot{x} + 4\dot{x} + 5x = 1$$

were the right and side is an input $r(t) = 1$ and initial conditions $x(0) = x_0 = 1$ and $\dot{x}(0) = \dot{x}_0 = -1$. Taking Laplace transform yields

$$s^2X(s) - sx_0 - \dot{x}_0 + 4(sX(s) - x_0) + 5X(s) = R(s)$$

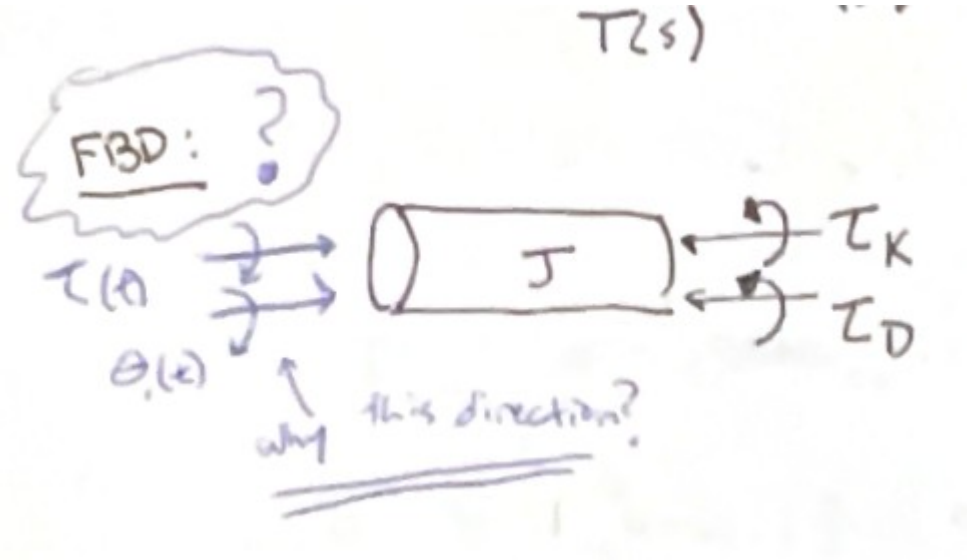
Solving for the output, $X(s)$, we see

$$X(s) = R(s)\left[\frac{1}{s^2 + 4s + 5}\right] + [(s+4)x_0 + \dot{x}_0]\left[\frac{1}{s^2 + 4s + 5}\right]$$

where the transfer function is

$$G(s) = \frac{1}{s^2 + 4s + 5}$$

Modeling and FBDs



USV Surge: First Order Response

b)

TF → $\frac{1}{36s + 18.75}$

$m\dot{u} + Du = T$

$T(0) = 37.5N$

$U(s) = \frac{1}{(36s + 18.75)} \frac{37.5}{s}$

$U(s) = \frac{37.5}{s(36s + 18.75)}$

residue $U(s) = \frac{-2}{s + 18.75} + \frac{2}{s}$ → $\mathcal{L}^{-1}$ → $-2(e^{-18.75t})$ + $2u(s)$

-0.5208

$u(t) = 2(1 - e^{-0.5208t})$

$U(t) = 2(1 - e^{-0.5208})$

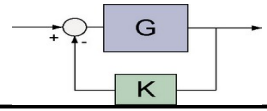
based on Matlab output how would I recognize that $e^{-0.5208t} = e^{t/\tau}$ where $\tau = M/D$?

$U(s)(Ms + D) = 37.5/s$

$U(s) = \left(\frac{1}{Ms + D}\right)\left(\frac{37.5}{s}\right)$? had to look this one up in the answerguide

$\mathcal{L}^{-1}[U(s)] = 2(1 - e^{t/\tau})$ where $\tau = M/D$

Inverse Laplace with Complex Poles



G(s)

$$G(s)=\frac{2}{s(s^2+s+16.25)}$$

```
num = 2;
den = [1 1 16.25 0];
[R2,P2,K2] = residue(num,den)
```

```
R2 = 3×1 complex
  -0.0615 + 0.0077i
  -0.0615 - 0.0077i
   0.1231 + 0.0000i
P2 = 3×1 complex
  -0.5000 + 4.0000i
  -0.5000 - 4.0000i
   0.0000 + 0.0000i
K2 =

     []
```

$$[K,p,k]=residue(num,den)$$

$$K=\begin{matrix}\alpha-\beta i\\ \alpha+\beta i\\ \dots\end{matrix}\qquad p=\begin{matrix}-a+\omega i\\ -a-\omega i\\ \dots\end{matrix}\qquad k=[\,]$$

$$F(s)=2\frac{(\alpha(s+a)+\beta\omega)}{(s+a)^2+\omega^2}+\dots$$

$$f(t)=2[\alpha e^{-at}\cos(\omega t)+\beta e^{-at}\sin(\omega t)]$$

$$a=0.5,\omega=4,\alpha=-0.0615,\beta=-0.0077$$

$$G(s)=2\left[\frac{-0.0615(s+0.5)-0.0077(4)}{(s+0.5)^2+4^2}\right]+\frac{0.1231}{s}$$

$$g(t)=2e^{-0.5t}[-0.0615cos(4t)-0.0077sin(4t)]+0.1231$$

Nise 2.12

Transfer functions, initial conditions and block diagrams

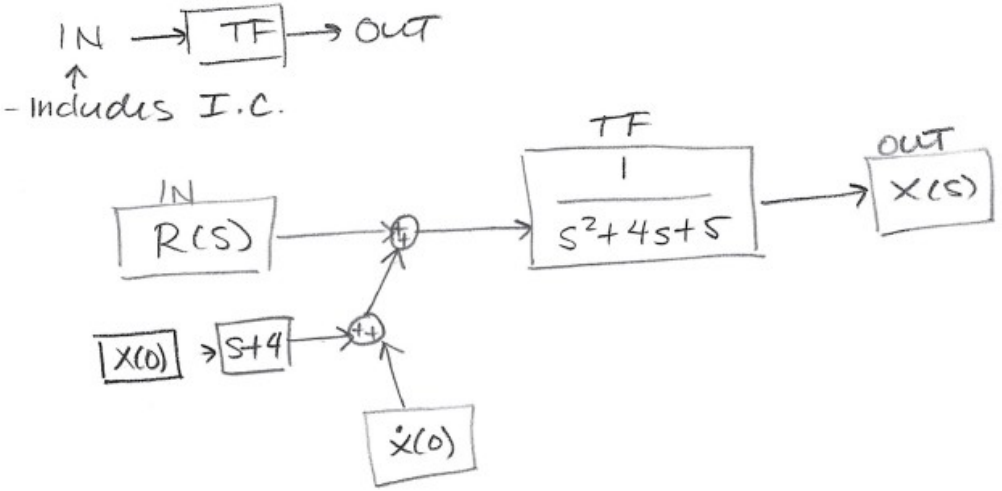
See solutions notes...

$$X(s) = \underbrace{\frac{R(s)}{s^2+4s+5}}_{\text{forced}} + \underbrace{\frac{s+3}{s^2+4s+5}}_{\text{free}}$$

$$G(s) = \frac{X(s)}{R(s)} \rightarrow \frac{1}{s^2+4s+5}$$

← why do we only focus on the forced response?

*Had no idea about how to draw block diagram with ICs



Solving ODEs with Laplace

a) $\frac{dx}{dt} + 7x = 5u(t) \qquad x(t=0) = 0$

$$sX(s) - \cancel{x(0)}^{0} + 7X(s) = \frac{5}{s}$$

$$X(s)(s+7) = \frac{5}{s}$$

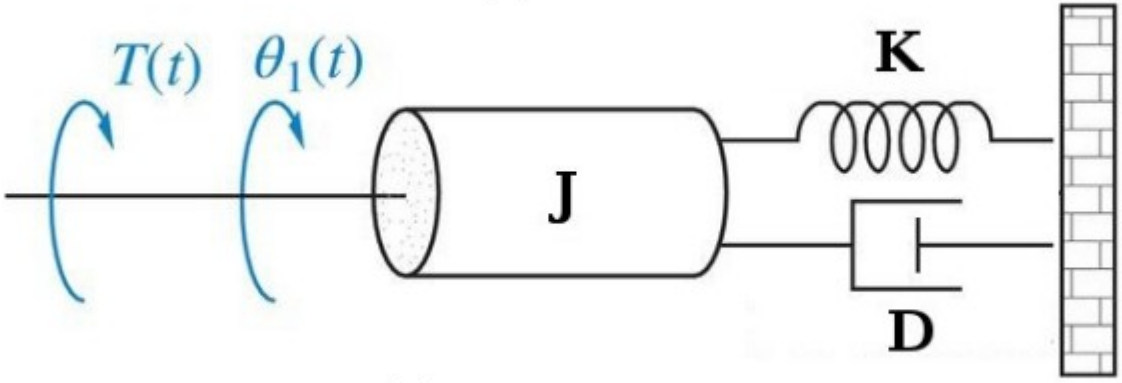
$$X(s) = \frac{5}{s(s+7)} = \frac{0.7143}{s} - \frac{0.7143}{s+7}$$

$$\mathcal{L}^{-1}\{X(s)\} = x(t) = [0.7143 - 0.7143e^{-7t}]u(t)$$

on an exam would the omission of $u(t)$ lose points?

One-DOF Rotational Model

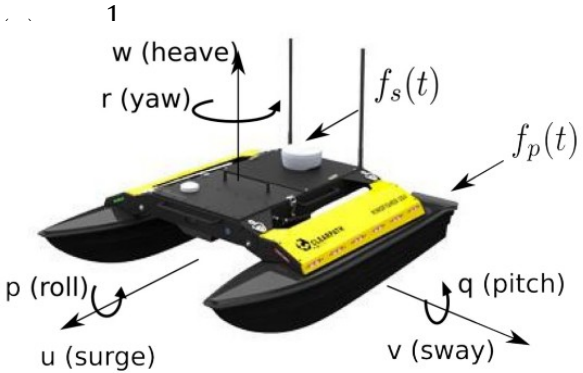
9) Find the transfer function $G(s)=\frac{\Theta_1(s)}{T(s)}$ for the system

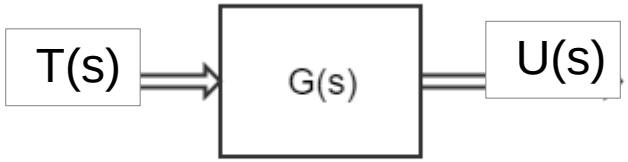


where T(s) is the input torque, $\theta_1(t)$ is the angular displacement and J, D, K are linear inertia, viscous damping and spring elements respectively.

I struggled setting this up.
I see the MOE about RHR
Is it RHR on right because of reaction?

USV Model – Foreshadow of this week



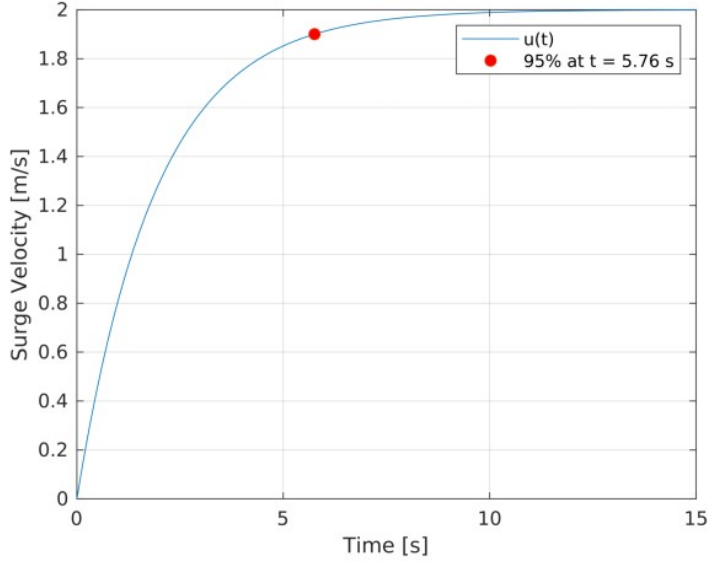


Laplace domain

$$G(s) = \frac{1}{Ms + D}$$

We will learn that τ is the *time constant* of this first order response: $\tau = M/D = 1.92s$.

From G(s) we can directly answer:
* What is the steady-state speed of the vessel?
* How long does it take for the vessel to reach 95% of the steady-state speed?



First-Order Time Response – The Long Way

10)

a) thrust → m 36 kg ← drag

b) ~~[illegible]~~

c)

D / drag = 18.75 N

T / thrust = 37.5 N

$m\dot{v} + Dv = T$

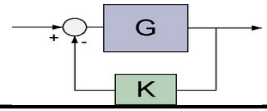
$$\frac{V(s)}{T(s)} = \frac{1}{sM + D}$$

d)

$$V(s) = \frac{.1}{5m + D} \cdot \frac{37.5}{s}$$

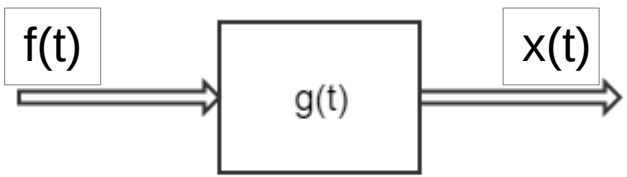
b) even looking at the answers, I am a bit confused ~~on~~ at this point on. I understand bringing it back to the time domain, but short of just copying, I do not arrive at that code.

Transfer Function

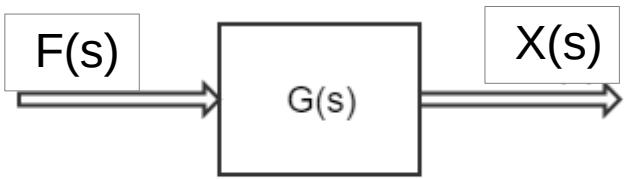


a) $\frac{X(s)}{F(s)} = \frac{7}{s^2+5s+10}$ ⇒ $s^2+5s+10\ X(s) = 7\ F(s)$

$\ddot{x} + 5\dot{x} + 10x = 7t$ is $F(s)$ the input function $f(t)$ or the step function



Time domain

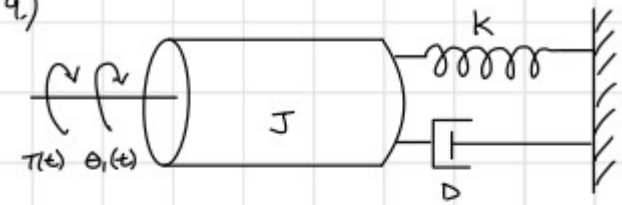


Laplace domain

Modeling



9.)



FBD:

R.H.R.



* CONFUSED BY DIRECTION OF ROTATION IN PROBLEM VS. DIRECTION IN SOLUTIONS FOR T(t) AND θ(t)

$$\Sigma T = J\ddot{\theta}_1 = T + T_k + T_D$$

$$\hookrightarrow T_k = k\theta_1$$

$$T_D = D\dot{\theta}_1$$

$$J\ddot{\theta}_1 = T(t) + k\theta_1 + D\dot{\theta}_1$$

$$\hookrightarrow T(t) = J\ddot{\theta}_1 - k\theta_1 - D\dot{\theta}_1$$

$$T(s) = (Js^2 - k - Ds)\,\theta(s)$$

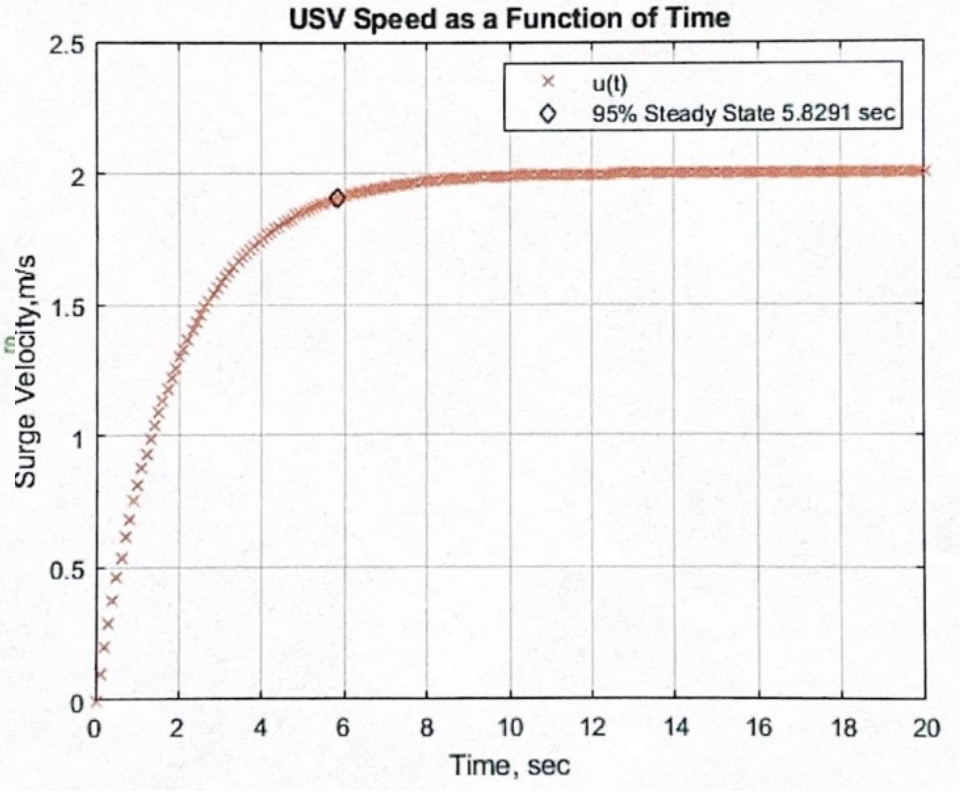
$$\frac{\theta_1(s)}{T(s)} = \frac{1}{Js^2 - Ds - k}$$

* SIGNS OFF PROBABLY

MATLAB E.g.,

```
t=linspace(0,20,200);
u_t=2.*(1-exp(-.5208.*t));
plot(t,u_t, 'rx');
title('USV Speed as a Function of Time')
ylabel('Surge Velocity,m/s')
xlabel('Time, sec')
ylim([0 2.5])
grid on;
hold on;

p95=find(u_t > (0.95*2),1); %find 95% and 1 is saying dis
 integer
x95=t(p95);
y95=u_t(p95);
plot(x95,y95,'kd')
legend('u(t)',['95% Steady State ' num2str(x95) ' sec'])
```

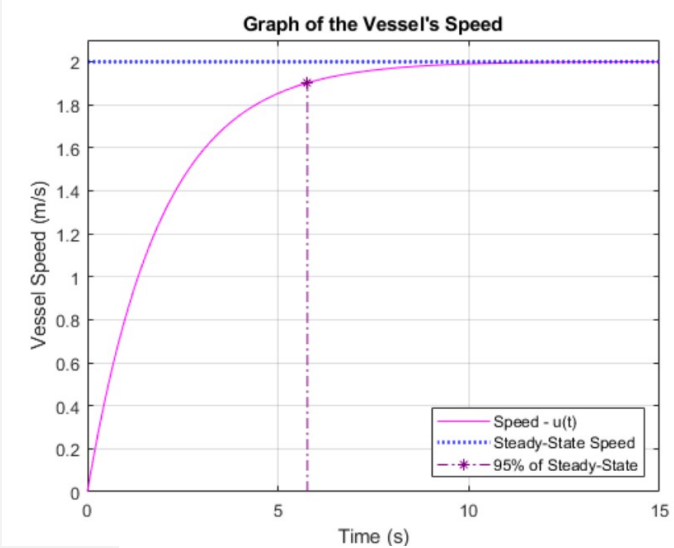


MATLAB – Symbolic toolbox

```
T = 37.5; M = 36; D = 18.75;
TransferFn=[1]/[s*(M*s+D)];
Us=TransferFn*T;
Ut=ilaplace(Us);
Time = linspace(0,15,1000);
Speed = double(subs(Ut,t,Time));
SS = double(limit(Ut,t,inf));
Index95 = find(Speed>(.95*SS),1);

figure(1); plot(Time,Speed,'m-'); hold on;
yline(SS,'b:','linewidth',2);
plot([Time(Index95), Time(Index95)],[-1, Speed(Index95)],'-.*','color',[.5 0 .5]);
ylim([0, 2.1]); grid on;
xlabel('Time (s)'); ylabel('Vessel Speed (m/s)'); title("Graph of the Vessel's Speed")
legend('Speed - u(t)','Steady-State Speed','95% of Steady-State','location','southeast');

fprintf('Expression of the vessel speed as a function of time \n'); disp(Ut)
fprintf('Steady-State Speed of the vessel is %1.2f m/s \n',SS);
fprintf('Time for the vessel to reach 95%% Steady-State is %1.4f seconds \n', Time(Index95));
```




```
Expression of the vessel speed as a function of time
2 - 2*exp(-(25*t)/48)

Steady-State Speed of the vessel is 2.00 m/s
Time for the vessel to reach 95% Steady-State is 5.7658 seconds
```

Sign error and notation



2c) h(t)'s last term should be (+)

Am I ok to use x' for $\frac{dx}{dt}$, x'' for $\frac{d^2x}{dt^2}$, etc?

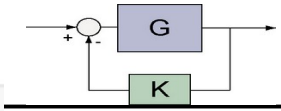
c) $H(s) = \frac{2(-0.0096)s}{s^2+9} + \frac{2(-0.0064)(3)}{s^2+9} + \frac{-0.0126}{s+3.7321} + \frac{0.0318}{s+0.2679}$

$= -0.0192\left(\frac{s}{s^2+9}\right) - 0.0128\left(\frac{3}{s^2+9}\right) + \text{"} + \text{"}$

$h(t) = -0.0192\cos(3t) - 0.0128\sin(3t) - 0.0126e^{-3.7321t} + 0.0318e^{-0.2679t}$

★ ANSWER KEY STATES LAST TERM IS NEGATIVE

H(s)



c) $$H(s)=\frac{1}{(s^2+9)(s^2+4s+1)}$$

```
num = 1;
den = conv([1 0 9],[1 4 1]);
[R3,P3,K3] = residue(num,den)
```

```
R3 = 4×1 complex
  -0.0126 + 0.0000i
  -0.0096 + 0.0064i
  -0.0096 - 0.0064i
   0.0318 + 0.0000i
P3 = 4×1 complex
  -3.7321 + 0.0000i
   0.0000 + 3.0000i
   0.0000 - 3.0000i
  -0.2679 + 0.0000i
K3 =

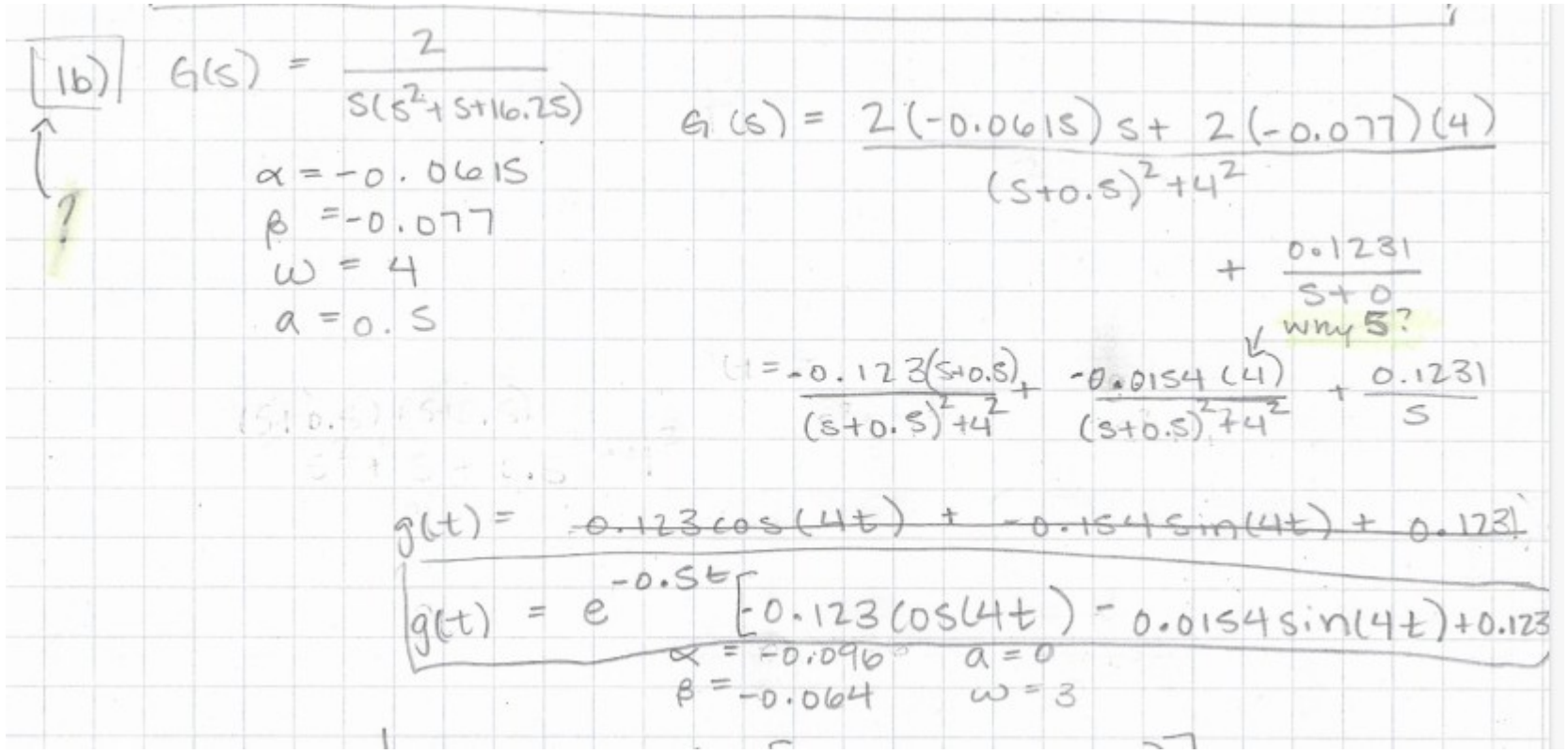
     []
```

$$a = 0, \omega = 3, \alpha = -0.0096, \beta = -0.0064$$

$$H(s) = 2\left[\frac{-0.0096(s) - 0.0064(3)}{s^2 + 3^2}\right] + \frac{-0.0126}{s + 3.7321} + \frac{0.0318}{s + 0.2679}$$

$$g(t) = 2[-0.096cos(3t) - 0.0064sin(3t)] - 0.0126e^{-3.7321t} - 0.0318e^{-0.2679t}$$

Inverse Laplace w/ Complex Roots



G(s)

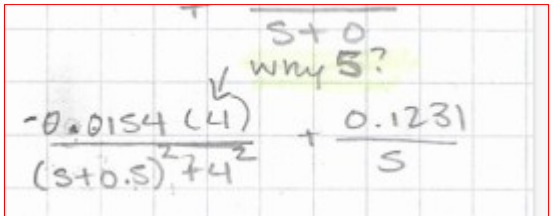
```
num = 2;
den = [1 1 16.25 0];
[R2,P2,K2] = residue(num,den)
```

```
R2 = 3×1 complex
  -0.0615 + 0.0077i
  -0.0615 - 0.0077i
   0.1231 + 0.0000i
P2 = 3×1 complex
  -0.5000 + 4.0000i
  -0.5000 - 4.0000i
   0.0000 + 0.0000i
K2 =
     []
```

$$a = 0.5, \omega = 4, \alpha = -0.0615, \beta = -0.0077$$

$$G(s) = 2\left[\frac{-0.0615(s+0.5) - 0.0077(5)}{(s+0.5)^2 + 4^2}\right] + \frac{0.1231}{s}$$

$$g(t) = 2e^{-0.5t}[-0.0615cos(4t) - 0.0077sin(4t)] + 0.1231$$



$$[K, p, k] = residue(num, den)$$

$$K = \begin{matrix} \alpha - \beta i \\ \alpha + \beta i \\ \dots \end{matrix}$$

$$p = \begin{matrix} -a + \omega i \\ -a - \omega i \\ \dots \end{matrix}$$

$$k = []$$

If we use this result directly we get a messy expression

$$F(s) = \frac{\alpha - \beta i}{s + (a - bi)} + \frac{\alpha + \beta i}{s + (a + bi)} + \dots$$

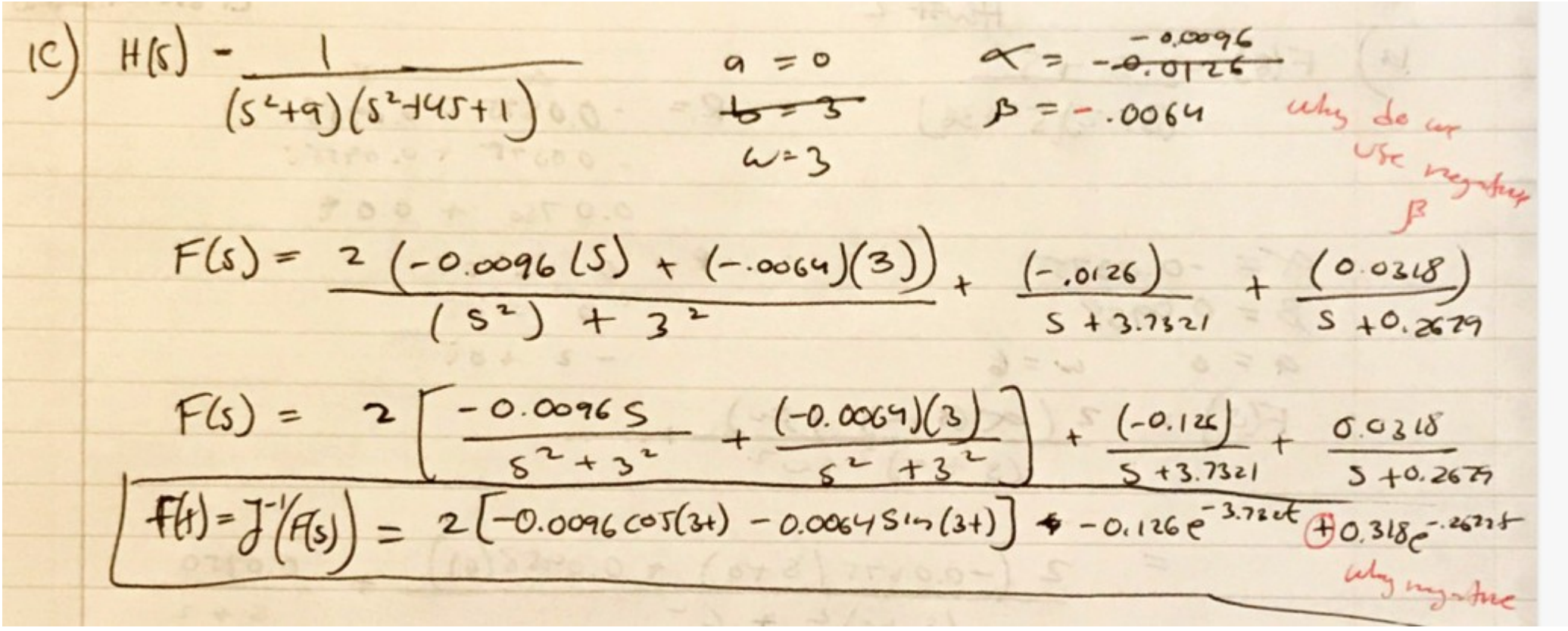
which we can simplify to

$$F(s) = 2\frac{(\alpha(s+a) + \beta\omega)}{(s+a)^2 + \omega^2} + \dots \quad .$$

We can then take the inverse Laplace transform of this term...

$$f(t) = 2[\alpha e^{-at}\cos(\omega t) + \beta e^{-at}\sin(\omega t)] \quad .$$

Inverse Laplace w/ Complex Roots



H(s)

```
num = 1;
den = conv([1 0 9],[1 4 1]);
[R3,P3,K3] = residue(num,den)
```

```
R3 = 4×1 complex
  -0.0126 + 0.0000i
  -0.0096 + 0.0064i
  -0.0096 - 0.0064i
   0.0318 + 0.0000i
P3 = 4×1 complex
  -3.7321 + 0.0000i
   0.0000 + 3.0000i
   0.0000 - 3.0000i
  -0.2679 + 0.0000i
K3 =
     []
```

$[K, p, k] = residue(num, den)$

$K =$

$\alpha - \beta i$

$\alpha + \beta i$

...

$p =$

$+\omega i$

$-\omega i$

...

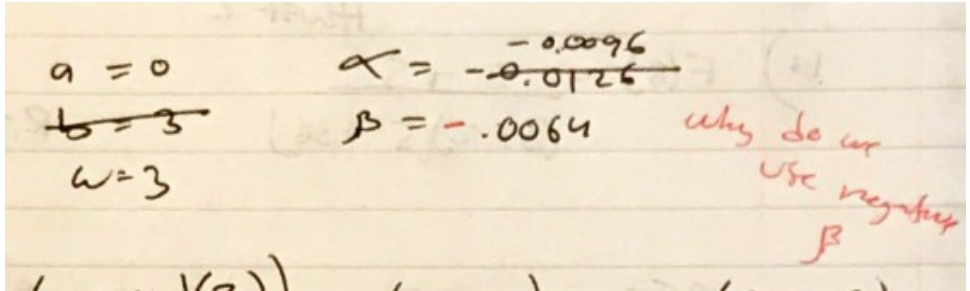
$k =$

[]

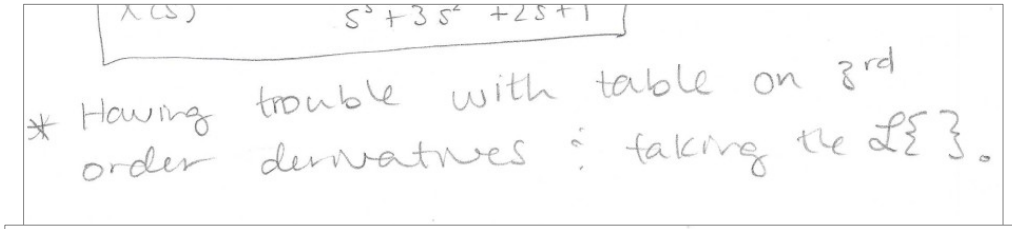
resulting expression for F(s)

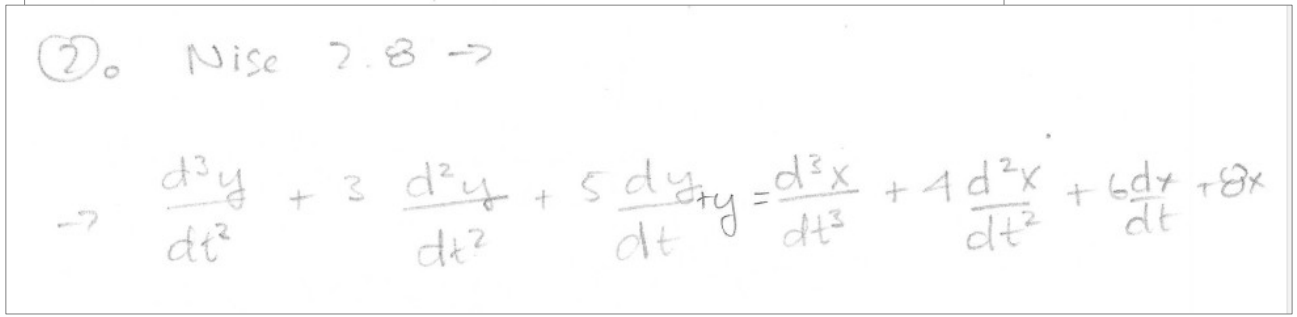
$$F(s) = \frac{2(\alpha)s + 2(\beta)(\omega)}{s^2 + \omega^2} + \ldots \quad .$$

$$a = 0, \omega = 3, \alpha = -0.0096, \beta = -0.0064$$



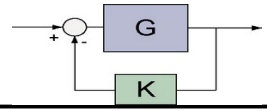
Higher order derivatives and TFs





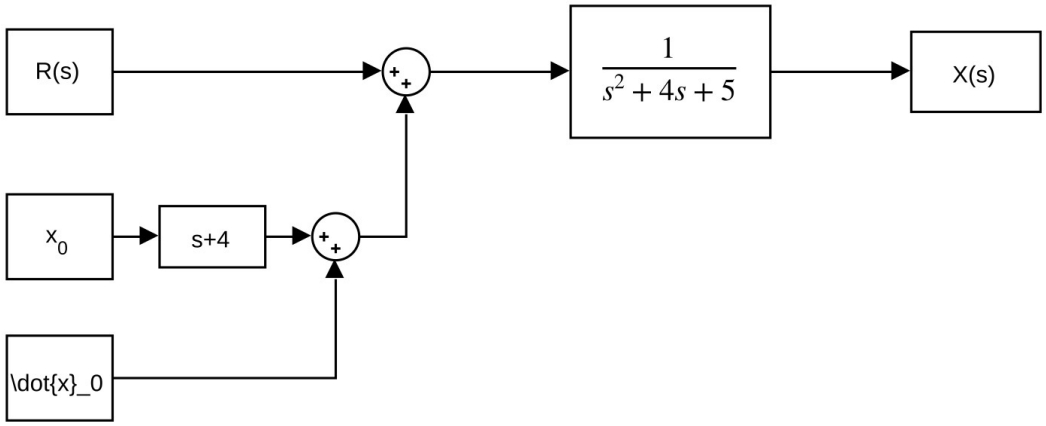
$$\mathcal{L}\left[\frac{d^n f}{dt^n}\right] = s^n F(s) - \sum_{k=1}^{n} s^{n-k} f^{k-1}(0-)$$

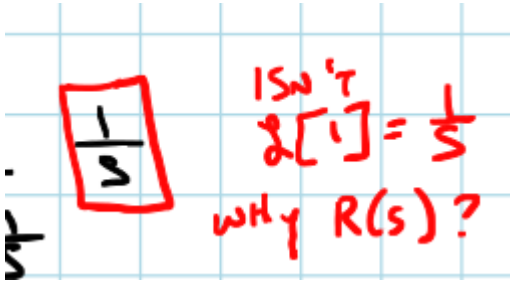
HW2Review



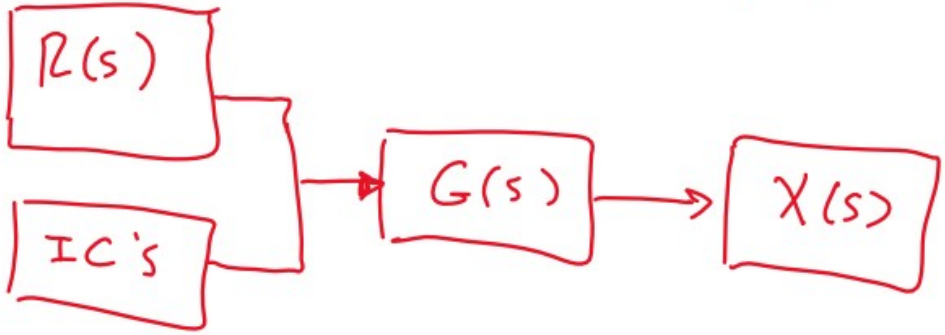
- Nise 2.12 – block diagram of w/ IC's

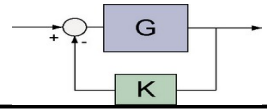
$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 1$$





I HAD TO LOOK @ SOLN TO DRAW BLOCK DIAGRAM BELOW

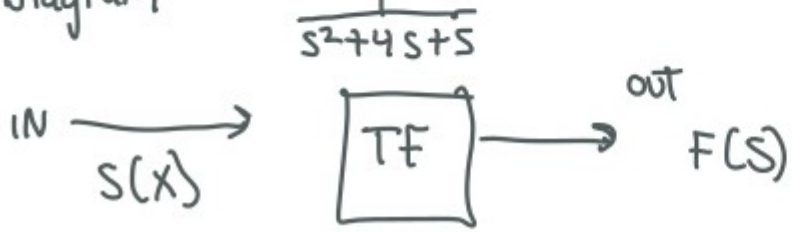




4) $12\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 1 \qquad x(0) = 1 \quad \dot{x}(0) = -1$

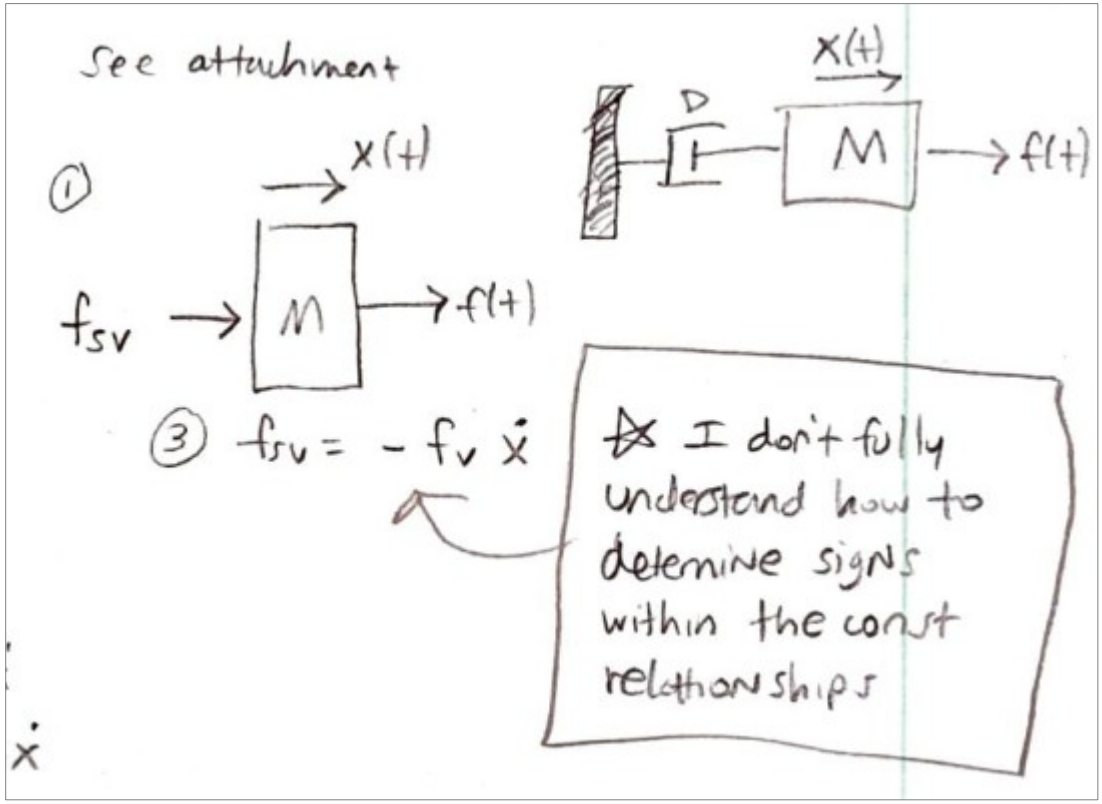
$$(s^2 + \dot{x}(0) + 4s + x(0) + 5)\, S(x) = F(s)$$

Block diagram

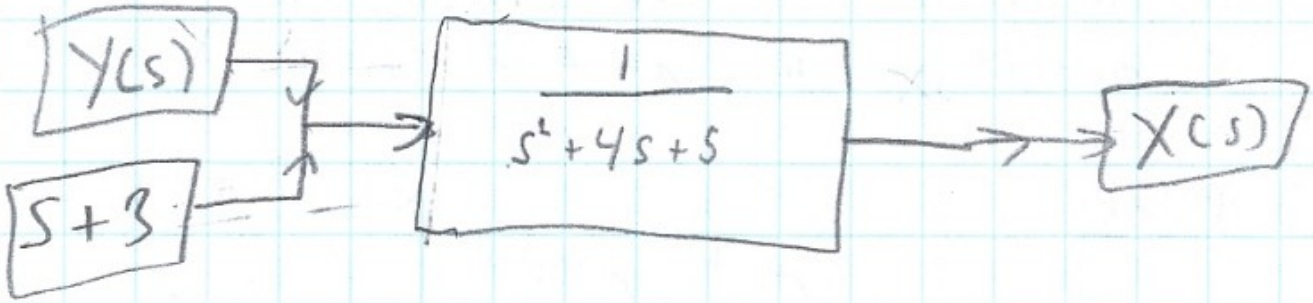


← not quite sure if this is correct. Is the S(x) the input?

Modeling - Translation



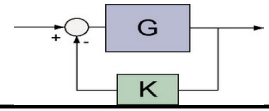
20-3



* After Watching the video I see where I can improve the diagram by using "+ +".

* Why is it important to leave the function in terms of X_o and X_o? Does it really matter? Same result, but a matter of formality?

Inverse Laplace w/ Complex Roots



a) $F(s) = \frac{s+5}{(s+2)(s^2+36)}$

num = [1 5]

den = conv([1 2], [1 0 36])

[R1 P1 k1] = residue(num, den)

$a = 0$

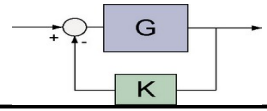
$w = 6$

$\alpha = -0.0375$

$\beta = 0.0958$

$F(s) = 2e^{at}\left[\frac{w(s+a)-\beta(5)}{(s+a)^2+w^2}\right] + \frac{0.1231}{s}$

Lost, I do not know what to do from here..

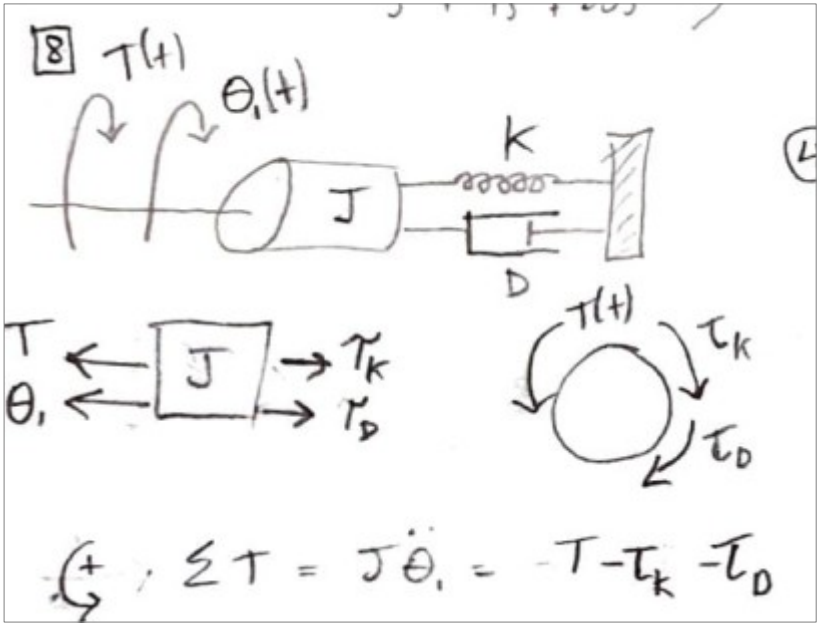


6) See MATLAB Scripts → *

why do root matrices need to be transposed to make the zpk function work?

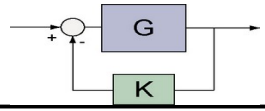
Receive hurtcat error otherwise.

Modeling - Rotation



★ do not understand how to determine direction of τ_K, τ_D sign

USV Model - Surge



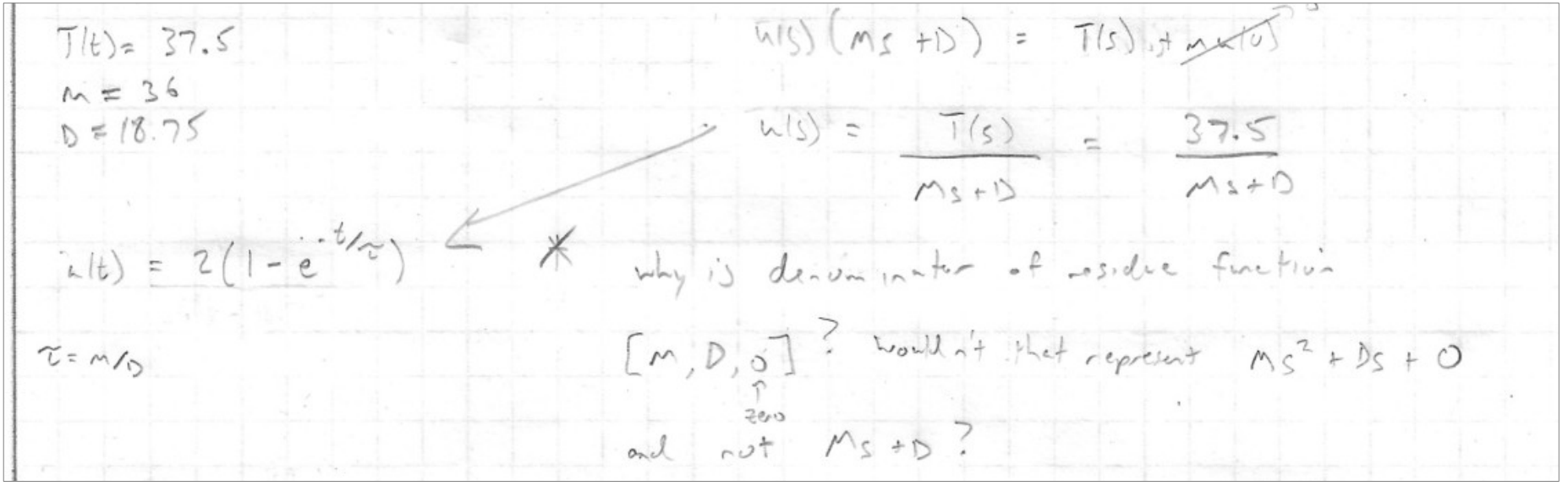
$$\frac{U(s)}{T(s)} = \frac{1}{ms + D}$$

* I reworked problem after seeing the solution. I had originally worked it starting at $m\ddot{x}$ (position) vice velocity.

* DON'T UNDERSTAND WHERE WE GET T is a time constant.

$T = M/D = 1.925$?

USV Step Response



USV Model

d) Find the transfer function for the yaw direction where the input is $\tau(t)$ and the output is the angular displacement $r(t)$.

$$\frac{R(s)}{\tau(s)} = \frac{1}{Is^2 - sD} + \frac{(s+1)r(0) + r'(0)}{Is^2 - sD}$$

0 IF IC's = 0

I DON'T REALLY GET WHY SAM SAYS: $\frac{1}{Is+d} = \frac{\dot{R}(s)}{\tau(s)}$

Let $y(t) = \dot{r}(t)$

$$I\dot{y} + dy = \tau(t)$$

$$\frac{Y(s)}{\tau(s)} = \frac{\dot{R}(s)}{\tau(s)} = \frac{1}{Is+d}$$